PORTFOLIO PROJECT PDF

Airbnb Listing Success Analysis
ALY6110 Module 6 Final Project

This is my shorter portfolio version of a Fall 2024 group project for ALY6110: Data Management and Big Data. We used New York City Airbnb open data to study what may be related to listing engagement. My main part was the later-stage analytics: text analysis on house rules, topic modeling, and review prediction.

R | ggplot2 | Tableau Dashboard | NLP | Random Forest

SIMPLE NOTE

This PDF rewrites my original report, slides, R script, and uploaded course materials into a shorter portfolio format. I kept the language simple and focused on what an HR reviewer or interviewer can understand quickly.

BUSINESS PROBLEM

What makes some Airbnb listings get more guest attention than others? I used number of reviews as a practical proxy for listing engagement, then checked how price, room type, location, stay rules, and house-rule text may relate to that outcome.

PROJECT INFO

At a glance

Course: ALY6110 - Data Management and Big Data

Term: Fall 2024

Type: Group project

Team: Cheng Liu, Wenzhuo Zhang

Original deliverables: Final report, slides, dashboard, R script

Dataset: NYC Airbnb Open Data from Kaggle

Rows / columns: 102,599 rows, 26 columns

Main tools: R, tidyverse, tidytext, topicmodels, randomForest, Tableau

Model note: Random forest to predict number of reviews from numeric and text-derived features

OVERVIEW

Project Snapshot

The original project combined data cleaning, descriptive analysis, dashboard work, text mining, and predictive modeling. This portfolio version highlights the full workflow, but puts more attention on the part I want to show most clearly: turning messy listing text and business-style Airbnb data into readable findings.

BUSINESS FOCUS

Listing engagement

Used review count as a simple demand proxy.

DATA SCALE

100K+ listings

NYC listings with price, location, rules, and review fields.

METHODS USED

EDA + NLP + ML

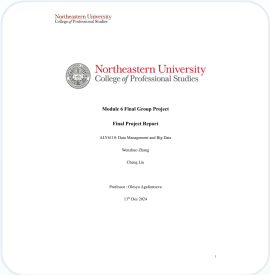
Cleaning, visuals, dashboarding, sentiment, LDA, random forest.

ORIGINAL DELIVERABLES

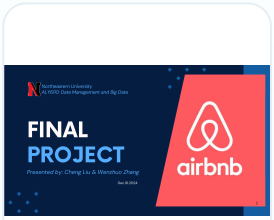
Report + slides

Also supported with dashboard views and R code.

Original source materials consolidated here



Final report PDF



Final presentation PDF

I reused the original report structure, slide visuals, and my R workflow. I also rebuilt a few charts from the CSV so the portfolio version is easier to read.

My role in the project

- Worked with the Airbnb dataset in R and helped with cleaning and early analysis setup.
- Tokenized house_rules text and built sentiment scores with tidytext.
- Ran LDA topic modeling and merged topic features back into the main table.
- Trained and interpreted a random forest model for number of reviews.

Contribution note

My clearest ownership was the later part of the project: textual insights, modeling, and explanation of model results. I also touched cleaning and clustering in code, even though the report split presentation work across both team members.

BUSINESS PROBLEM AND ANALYSIS PLAN

Problem, Dataset, and Analysis Plan

I treated this as a practical analytics task: understand the Airbnb market, then test which listing features may be related to stronger engagement. The goal was not to claim perfect prediction, but to find a useful decision pattern from a large real-world dataset.

PROJECT PURPOSE

Show a full big-data-style workflow on public data: clean the table, check market patterns, build an interactive view, extract text features from house rules, and test whether those features add signal when predicting reviews.

Dataset summary	Details
Main table	NYC Airbnb Open Data from Kaggle
Second table	No second source table; text fields came from house_rules inside the same file
Merge key	Listing id
Working data	102,599 rows x 26 columns raw; duplicates and missing price values were cleaned before analysis
Missing values	Numeric NA filled with mean, categorical NA filled with mode in the R workflow

STAKEHOLDERS

Airbnb hosts, small property investors, and anyone trying to understand where listing demand may be stronger in New York City.

MAIN QUESTIONS

Which listing features are related to more reviews?
Which neighborhoods and room types stand out?
What themes appear in house rules?
Can text features improve a review prediction model?

MY WORKFLOW

Load -> clean -> engineer features -> EDA -> dashboard -> text mining -> random forest -> interpretation -> summary.

SELECTED CODE - DATA SETUP

```
airbnb_data$price <- as.numeric(gsub("\\$", "",
airbnb_data$price))

airbnb_data$last.review[airbnb_data$last.review == ""] <-
NA
airbnb_data$last.review <- as.Date(airbnb_data$last.review,
format = "%m/%d/%Y")

airbnb_data[numerical_cols] <- lapply(
  airbnb_data[numerical_cols],
  function(x) ifelse(is.na(x), mean(x, na.rm = TRUE), x)
)

airbnb_data <- airbnb_data %>%
  mutate(
    price_per_night = price / minimum.nights,
    reviews_per_year = reviews.per.month * 12
  )
```

This was the basic preparation step. I standardized price, handled missing values, converted dates, and created a few simple features before analysis.

LAB / CODE NOTE

I mainly used R for cleaning, visual checks, text mining, and modeling. The project outputs were then organized into a written report, presentation, and dashboard-style view.

EXAMPLE FIELDS I USED

price, minimum nights, number of reviews, reviews per month

room type, neighbourhood group, neighbourhood, availability 365

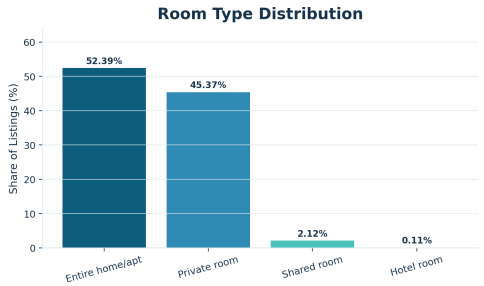
review rate number, lat, long, house_rules, id

EXPLORATORY DATA ANALYSIS

Key Findings and Visual Evidence

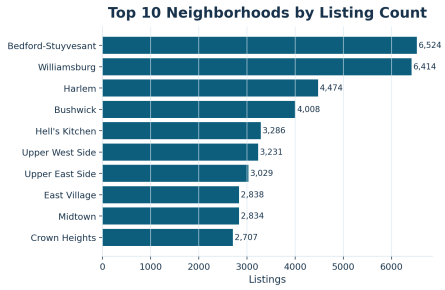
This part gave me the clearest market story. Before modeling, I first needed a practical read on supply mix, popular areas, and what a stakeholder would notice right away.

Room type mix



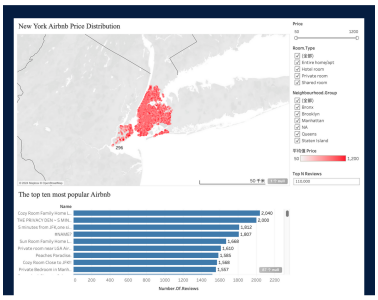
Entire home/apt and private room dominated the platform. Hotel room supply was tiny.

Neighborhood concentration



Brooklyn and Manhattan neighborhoods showed the densest supply, with Bedford-Stuyvesant and Williamsburg near the top.

Dashboard view from the original project



I kept the original dashboard view here because it shows the map, price filter, room-type filter, and top reviewed listings in one place.

WHAT THESE VISUALS TOLD ME

- The market was mostly non-hotel inventory, so guest choice was driven by private hosts and apartments.
- High listing concentration in Brooklyn and Manhattan suggested stronger competition but also stronger demand visibility.
- Top reviewed listings were not only about location; pricing and stay-policy choices also mattered.
- The dashboard made the project easier to explain to non-technical viewers because filters linked geography, budget, and listing popularity.

Quick business read

A good Airbnb listing strategy is not only about being in a popular place. It is a combination of location, room type, pricing, and how easy the stay rules feel to guests.

METHODS OR MODELING NOTE

Text Mining and Modeling Note

This was the part I most wanted to show in my portfolio. I took unstructured house-rule text, turned it into usable features, and combined those features with numeric listing fields in one prediction workflow.

SELECTED CODE SNIPPET

```
tidy_rules <- airbnb_data %>%
  select(id, house_rules) %>%
  unnest_tokens(word, house_rules) %>%
  anti_join(stop_words, by = "word")

rules_sentiment <- tidy_rules %>%
  inner_join(get_sentiments("bing"), by = "word") %>%
  group_by(id) %>%
  summarize(sentiment_score = sum(ifelse(sentiment ==
"positive", 1, -1)))

lda_model <- LDA(rules_dtm, k = 3, control = list(seed =
123))

rf_model <- randomForest(
  number.of.reviews ~ sentiment_score + Topic1 + Topic2 +
Topic3 +
    price + availability.365 + minimum.nights + room.type,
  data = train_data, ntree = 100, mtry = 3, importance =
TRUE
)
```

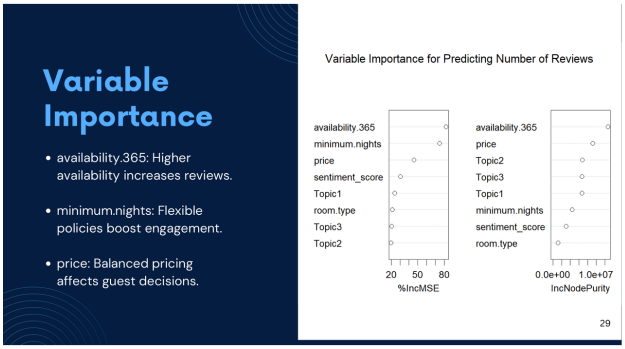
I used tidytext for sentiment, LDA for topic extraction, and random forest for a mixed-feature model.

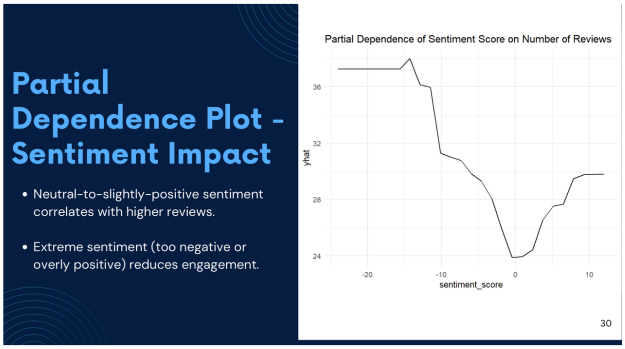
INTERPRETATION

- House rules were tokenized and scored with a simple lexicon-based sentiment approach.
- LDA reduced long rule text into three topic-style features that could be merged back by listing id.
- The final model mixed text-derived features with price, availability, minimum nights, and room type.
- The original project reported about MAE 19.18, RMSE 35.36, and R-squared 0.35, which was useful for pattern finding but not a perfect forecast.

The report also noted that availability.365, minimum.nights, and price were among the most influential features.

Model interpretation from the original project





The original slides summarized variable importance and a partial dependence view for sentiment_score. The main message was simple: availability and minimum-night rules mattered more than text alone, but text still added a readable signal.

EXECUTIVE SUMMARY

Conclusion, Recommendations, and Interview Use

The value of this project was not only the final model. It showed that I can move from messy public data to a practical story: clean the table, check the market structure, add text features, and explain what the results really mean without over-claiming.

MAIN FINDINGS

Supply was heavily concentrated in entire homes/apartments and private rooms.

Location, availability, minimum-night rules, and price were more important than any single text feature by itself.

RESULT MEANING

Listing engagement looked like a market-positioning problem. Hosts need a good balance of practical stay rules, visible availability, and pricing that fits the neighborhood and room type.

RECOMMENDATIONS

Keep pricing competitive, avoid overly restrictive minimum-night policies when possible, keep rules clear and neutral, and use dashboard views to compare listing strategy by area.

WHAT THIS PROJECT SHOWS ABOUT MY WORK

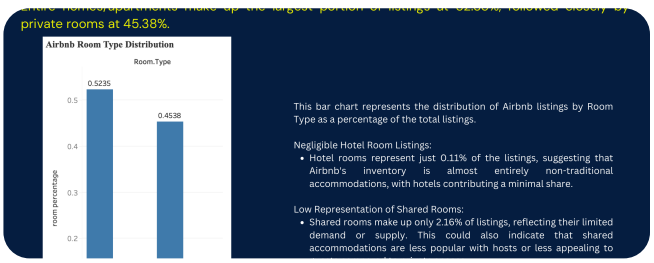
- I can clean large real-world data and organize a usable workflow.
- I can connect EDA, dashboard thinking, text mining, and model interpretation in one project.
- I can explain limitations honestly instead of overselling a model.
- I can work inside a group project while still owning a clear technical section.

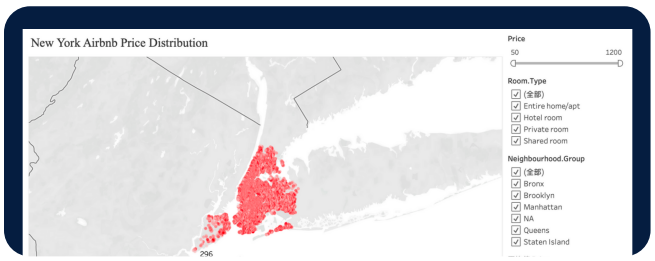
HOW I WOULD EXPLAIN THIS IN AN INTERVIEW

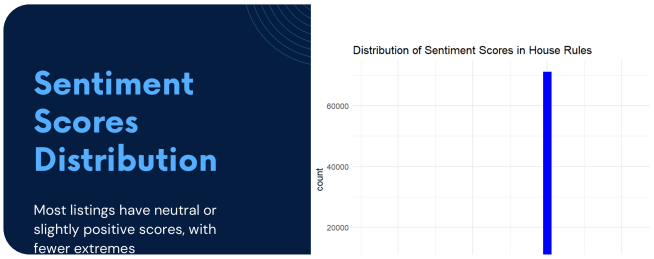
Simple version: This was our final project for a big data course. We analyzed 100K+ NYC Airbnb listings, and my main part was turning house-rule text into sentiment and topic features, then using those features with numeric listing data in a random forest model. The project showed me how to go from messy data to a business-style story with charts, dashboard views, and model interpretation.

> **Final note**
> I would present this project as an end-to-end analytics case, with my strongest contribution in textual insights and predictive modeling, while staying clear that it was a group project and not a solo build.

Quick Visual Gallery







I kept a small gallery at the end so a recruiter or interviewer can scan the style of work quickly: EDA, dashboard, sentiment, and summary visuals.